Shiv Nadar University
Department of Computer Science & Engineering

Data Structures
Course Code: CSD201

Lesson Plan
Academic Year: 2018 – 2019

Handled By:
- Pooja Malik
- C. Aravindan (Mentor)

Lecture Hours:
- Every Monday, Wednesday, and Friday, 09:00 AM – 10:00 AM
- additional lectures may be scheduled on other days if needed

Office Hours:
- Tuesday (02:00 PM - 04:00 PM)
- Wednesday (02:00 PM - 04:00 PM)
- By Email Appointment

Teaching Assistant:
- Surbhi Jain (surbhi.jain@snu.edu.in)

Text Book:
- Mark Allen Weiss, "Data Structures and Algorithm Analysis in C", Second Edition, Pearson, 1997. ISBN: 978-81-7758-358-8

Other Readings:
- Lecture notes and slides
- Reema Thareja, "Data Structures Using C", Second Edition, Oxford University Press, 2014.
- Y. Langsam, M. J. Augenstein, and A. M. Tanenbaum, "Data Structures Using C and C++", Second Edition, Pearson, 2015.
- E. Balagurusamy, "Data Structures Using C", McGraw-Hill, 2017.
- Yashavant P. Kanetkar, "Data Structures Through C", Second Edition, BPB Publications, 2003.
- ISRD Group, "Data Structures Using C", Tata McGraw-Hill, 2006.

Prerequisites:
- Introduction to Computing and Programming (Course Code CSD-101)

Lecture Schedule:

1) Introduction and Motivation
2) Linear Structures: Array, Sequence, Stack, and Queue
3) Analysis of Algorithms I (Asymptotic Notations)
4) Analysis of Algorithms II (Recurrence Relations)
5) Case Study: Quick Sort and Merge Sort
6) ADT List: Array implementation
7) Linked Lists
8) Variations of Linked Lists
9) Case Study: Polynomials
10) Case Study: Radix sort
11) Stack ADT implementations
12) Case Study: Postfix expressions
13) Queue ADT implementations
14) Tree Structures: Basic concepts and traversals
15) Case Study: Huffman Codes
16) Revision-I
17) Binary Trees
18) Binary Search Trees I
19) Binary Search Trees II
20) Threaded BST
21) AVL Trees I
22) AVL Trees II
23) Splay Trees
24) B+ Trees I
25) B+ Trees II
26) Priority Queues
27) Case Study: Heap Sort
28) Randomized Algorithms: Quick Sort
29) Disjoint Set ADT: Basic implementation
30) Disjoint Set ADT: Smart Union and Path Compression
31) REVISION: Tree Structures
32) Hashing: Hash functions and separate chaining
33) Hashing: Open addressing
34) Hashing: Rehashing and extensible hashing
35) Graph Structures
36) Directed Acyclic Graphs: Topological sort
37) Unweighted Shortest Paths
38) Weighted Shortest Paths (Dijkstra's Algorithm)
39) Minimum Spanning Trees: Prim's Algorithm
40) Minimum Spanning Trees: Kruskal's Algorithm
41) Depth-first search in undirected graphs
42) Case Study: Euler Circuits
43) Depth-first search in directed graphs
44) REVISION: Graph Structures
45) ADT and Object-Oriented Programming

Lab Schedule

Week 1
Rational ADT

Week 2
Empirical Study of Sorting Algorithms (Bubble & Insertion)

Week 3
Implementation of Quick and Merge Sort using recursion

Week 4
Implement Doubly and Circular Linked List (Applications will be given as Assignments)

Week 5
Graded Lab 1

Week 6

Implement and use Stack data type. Apply Stack to reverse a list of items. Avoid redundancy (by marking items) when building Stacks.

Week 7
Implement and use Queue Data Type. Re-implement the Queue data type with Doubly Linked Lists; Compare its use with the simple Queue.

Week 8
Graded Lab 2

Week 9
Implement Binary Tree, its traversals and different operations.

Week 10
Implement Binary Search Tree

Week 11
Implement Dijkstra's algortihm

Week 12
Implement MST Graph Algorithms

Week 13
Hashing (Open Addressing/Chaining)

Week 14
Graded Lab 3

Grading Rubrics:

- Class Participation 07%
- Assignments/Quizzes 25%
- Mid-term Exam 20%
- End-term Exam 30%
- Graded Lab Sessions 18%